



USER MANUAL - CODRUG

01. DESCRIPTION:

This is a tool applied to QSAR analysis using machine learning (ML) models. Its design includes everything from the preparation of labeled internal datasets, preprocessing, exploratory and statistical analysis, generation of molecular descriptors and feature engineering, construction and validation of regression models, classification and clustering, cross-validation, hyperparameter optimization, and prediction of classes or bioactivities in external databases. This interface guides users through complex processes, reducing the need for programming knowledge and IDE use. It was developed in Python 3.10.12 using the PyQt5/Qt 5.15.14 framework, with an intuitive and modular graphical interface organized into sequential tabs. The main libraries integrated into this tool were ChEMBL Web Client, Pandas, Numpy, Matplotlib, RDKit, PaDEL-Descriptor, TensorFlow, PyTorch, Pycaret, and Scikitlearn.

02. INSTALLATION:

2.1 Download the compressed file from the link:

<https://github.com/moimaian/CODRUG/archive/refs/heads/main.zip>

2.2 Extract the file to a working folder, for example, "\$HOME/CODRUG":

This can be done graphically in a file manager such as Nemo

- Extract the contents of the CODRUG-main.zip file to your system's home folder, creating the folder **\$HOME/CODRUG** (remove -main from the name)

It is also possible to extract and rename the folder by opening a terminal window in the Downloads folder and executing the commands:

```
sudo apt-get install unzip # If you have not installed
unzip CODRUG-main.zip
mv $HOME/Downloads/CODRUG-main $HOME/CODRUG
```

2.3 Still in the terminal, run the CODRUG.py file using the command:

```
cd $HOME/CODRUG && python3 ./CODRUG.py
```

Use the following command if you do not have python3 on your system:

```
sudo apt update
sudo apt install python3 -y
```

A window will then appear describing the requirements that are not installed in the environment so that the user knows what needs to be installed beforehand. This installation can be done through the interface itself, which will open automatically, or with the module_requirements.py file in the MODULES folder.

In this first run, the Python virtual environment will be created in **\$HOME/.venv/CODRUG**, where the program packages that are dependencies for running CODRUG will be installed (see topic 3).

For installations via the terminal, this environment can be activated with the command:

```
source $HOME/.venv/CODRUG/bin/activate
```



03. PREREQUISITES:

On the first startup, either automatically or through the HOME tab in the interface, you can open the prerequisite installation window. In the case of the HOME tab (**Figure 1**), there is an **Install requirements** option at the bottom that, when clicked, takes you to this installation window (**Figure 2**), where the prerequisites for running CODRUG and their versions will be listed. This is a list with checkboxes that allow the user to choose which programs they want to install and their versions. If this is your first time running the program, check all options. The proposed versions have been tested and are compatible with each other; however, if the user wishes, they can change to more recent versions. It is important to be careful when changing the versions of a package, as you need to make sure that it is compatible with the others. The predefined versions have already been tested and do not present any conflicts.

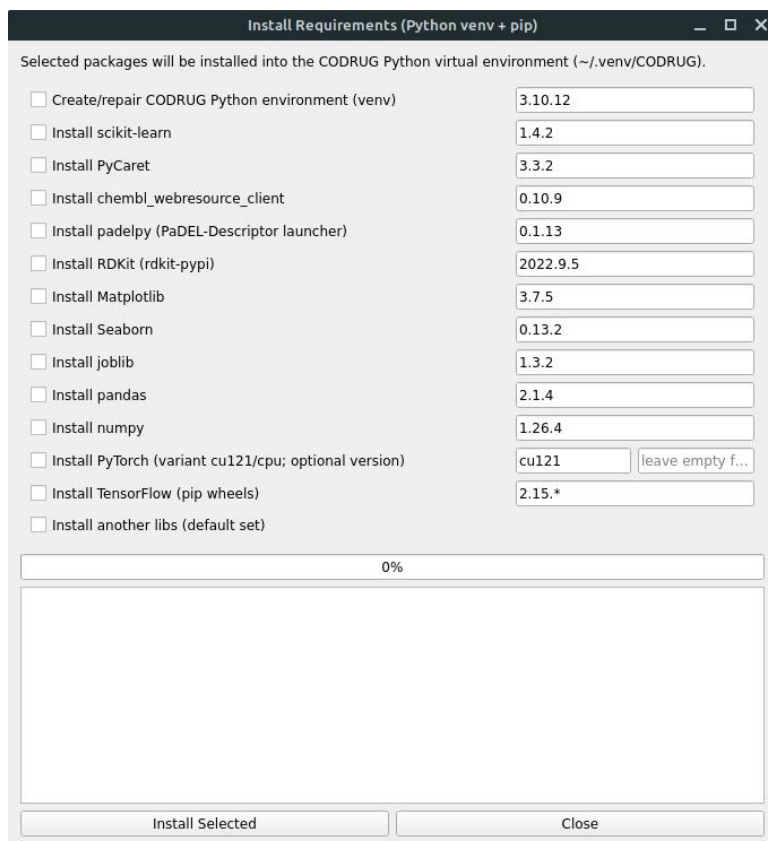


Figure 2. Configuration tab.

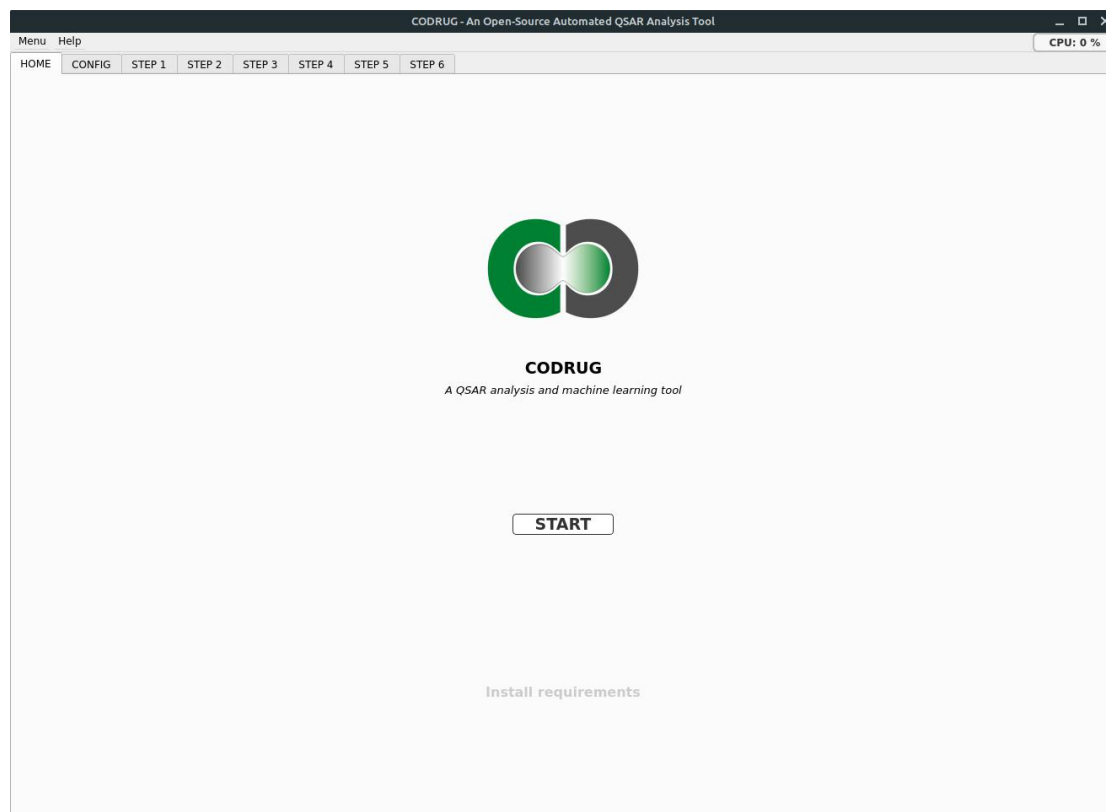


Figure 1. CODRUG HOME window.



04. DIRECTORY ORGANIZATION:

During the CODRUG startup process, the necessary directories are checked. If these directories are missing, they will be created in the folder where the **CODRUG.py** executable file is located. The organization of directories and subdirectories is described in **Figure 3** below, with some illustrative examples. It is worth noting that the USI is an identification code generated by Pycaret for each training/testing setup, making it possible to test various combinations of methods for scaling, selecting, or projecting data within the same project/job.

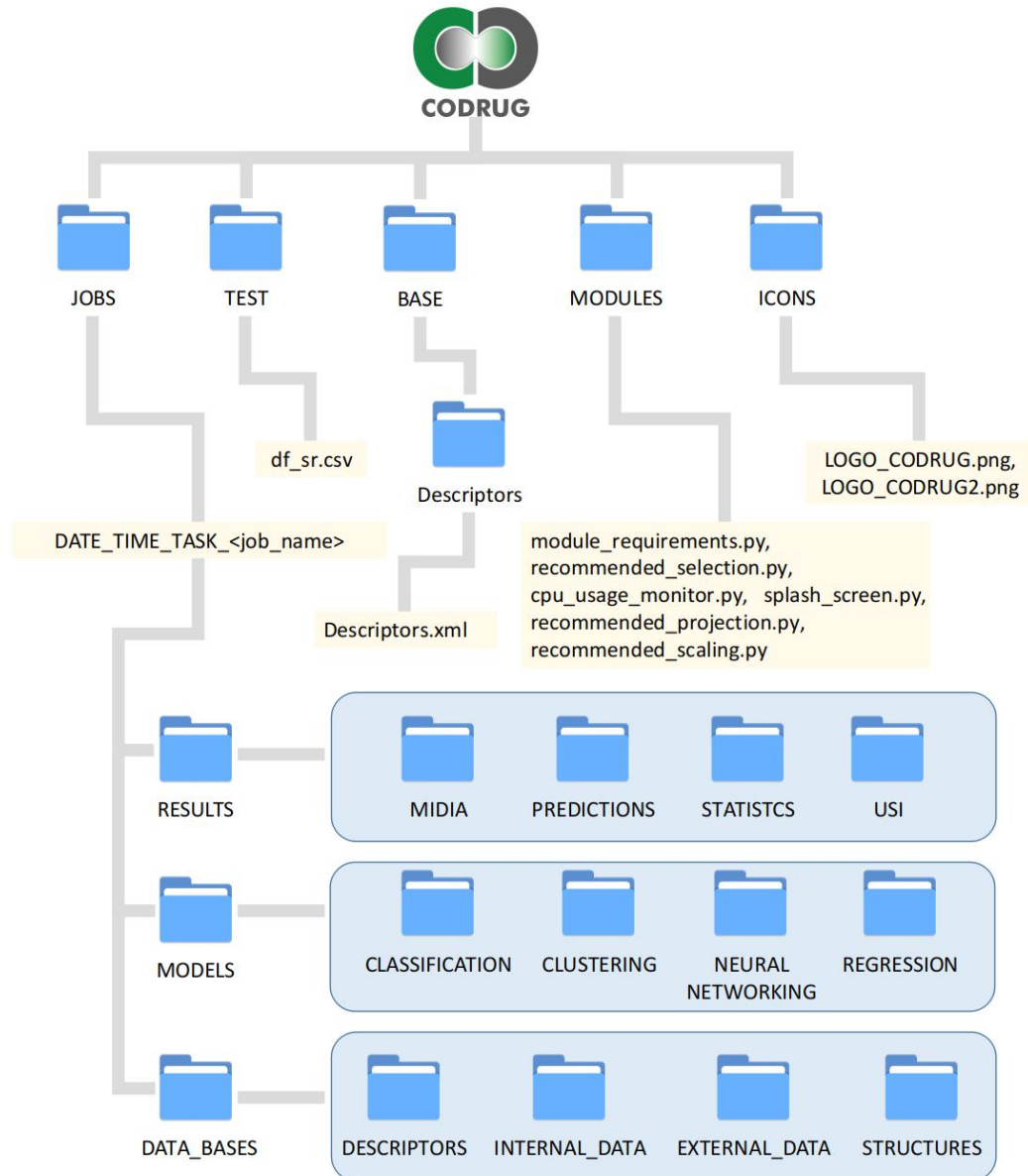


Figure 3. Directory organization.



05. GENERAL USAGE GUIDELINES:

After the **home tab**, the user must enter the project **settings tab** and select the **type of task** to be performed: **Classification**, **Regression**, **Clustering**. A new project must be created or a previous existing project must be resumed. Clicking the **Set Run Folder** button creates the necessary folders and subfolders (**Figure 4**).

Figure 4. Configuration tab.

Step 1 - Database preparation: In this step, the **ChEMBL Web Client API** is used to access the ChEMBL database. Several tools for searching and filtering data are available (**Figure 5**). The search begins by selecting the **Target Type** (*Organism, Single protein, Protein Complex, Cell-Line*), **Organism Name**, **Pref. Name**, and **Target ChEMBL ID**. When you fill in the name of the organism and click **Explore by target**, a new window with a dataset will open. From there, you can select the Target ChEMBL ID according to the value of num_compounds_tested. Only the **Pref. name** is not mandatory, and thus, with the other fields filled in, clicking on **Generate Dataset by activity** will generate the complete dataframe, saved in .csv format in the folder ~/CODRUG/JOBS/DATE_TIME_TASK_<JOB_NAME>/DATA_BASES/INTERNAL_DATA with the name df1_by_activity_<Target ChEMBL ID>_<Organism_name>.csv. This dataframe will serve as the basis for subsequent filtering with local access, avoiding repeated and time-consuming requests to the online database (**Figure 5**). For tasks involving cell line data, it is important to use the **Explore by cell-line** feature, filling in the fields cell_chembl_id, cell_name, cell_source_tissue, and cell_description to obtain the most appropriate base dataframe. This information can be found in advance on the web portal <https://www.ebi.ac.uk/chembl/>. Using the tools available in **View frequency graph**, it is possible to explore the data in each column, evaluating, for example, the Assay Type or Unit, thus allowing the most frequent options to be chosen (**Figure 6**).



CODRUG - An Open-Source Automated QSAR Analysis Tool

Menu Help CPU: 1 %

HOME CONFIG STEP 1 STEP 2 STEP 3 STEP 4 STEP 5 STEP 6

Dataset Preparation

Search Internal Data

Use ChEMBL Data

Target Type:
ORGANISM

Organism Name:
Mycobacterium tuberculosis

Pref. Name:
Ex.: DNA Gyrase / ATP Synthase...

Target ChEMBL ID:
CHEMBL360

Explore by target

Generate Dataset by activity

cell_chembl_id:
Ex.: CHEMBL3307651

cell_name:
Ex.: A549 / HepG2 / MCF7...

cell_source_tissue:
Ex.: Lung / Breast / Prostate...

cell_description:
Ex.: A549 (Lung carcinoma cells)

Explore by Cell-line

Assay Type:
F - Functional

Assay Metric:
Inhibition
pMIC
MIC
MIC50
MIC90
MBC
ID50

Assay Unit:
%
ug ml-1
uM
mg l-1
uM l-1
log CFU
mmol

Assay Strain:
Ex.: H37Rv

Assay ChEMBL ID:
Ex.: CHEMBL4665544

Assay Included Terms:
Case-insensitive/Contains, not exact - Ex.: resazurin, rema, alamar blue, maba, alamarblue, aba

Assay Excluded Terms:
Case-insensitive/Contains, not exact - Ex.: cornell, heat shock

Explore by assay

Assay description count

Molecule ChEMBL ID:
CHEMBL1234567

Molecule Name:
Metronidazole

Canonical SMILES:
CC(=O)OCl=CC=CC=C1C(=O)O

Activity ChEMBL ID:
CHEMBL4665544

Explore by molecule

Generate Base Dataset

View frequency graphs:

Internal Dataset list:
df1_by_activity_CHEMBL360_Mycobacterium tuberculosis.csv

Columns:
activity_id

Update

View Graph

<< BACK NEXT >>

Figure 5. Window for step 1 of dataset preparation.

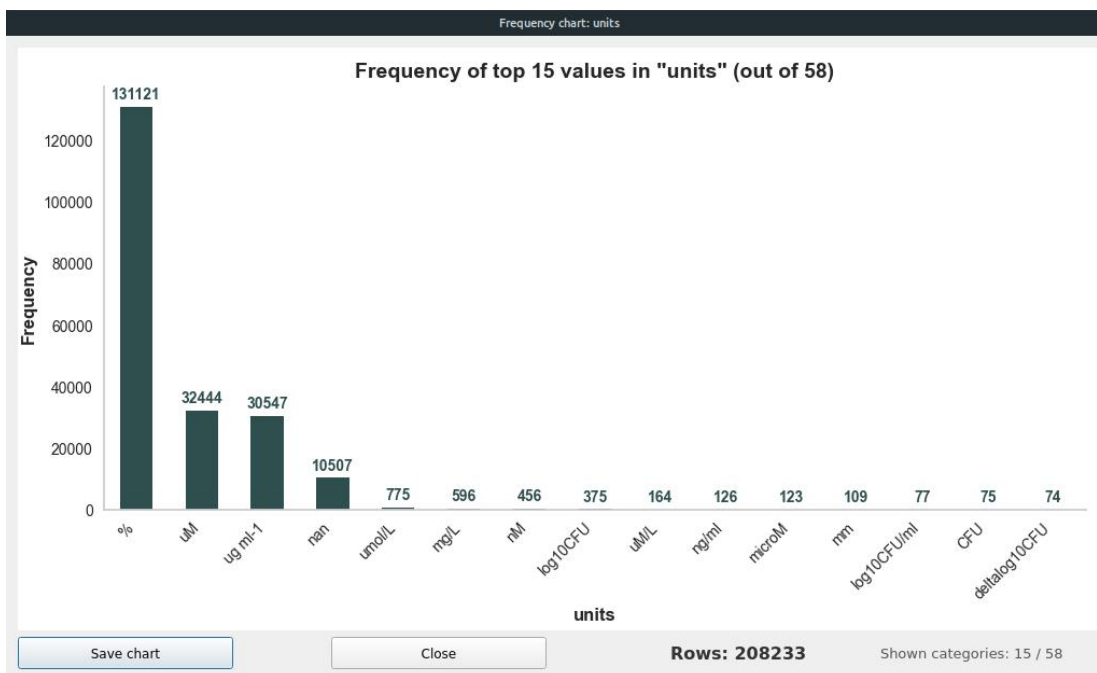


Figure 6. Distribution of data according to the type of unit of measurement.

The type of assay can be chosen from: **Functional (F)**, **Binding (B)**, **Pharmacokinetics (A)**, **Toxicity (T)**, or **Unsigned (U)**. By clicking on the **Assay description count** button, it is possible to check the different descriptions of the assays and their frequency in the dataframe, in addition to updating the lists of options in **Assay Metric** and **Assay Unit**. Multiple units can be chosen as long as they can be converted between each other based on the molar mass of



the compounds. The interconvertible concentration units are: *M*, *mM*, *uM*, *nM*, *mol/L*, *moll-1*, *mmol/L*, *mmoll-1*, *umol*, *umol/L*, *umoll-1*, μmol , $\mu\text{mol/L}$, μmol , $\mu\text{mol/L}$, *nmol*, *nmol/L*, *nmoll-1*, *ng/uL*, *ng/mL*, *ng/L*, *ug/uL*, *ug/mL*, *ug/L*, *mg/uL*, *mg/mL*, *mg/L*, *g/uL*, *g/mL*, *g/L*. And also the dose/weight units: *ng/g*, *ug/g*, *mg/g*, *g/g*, *ng/kg*, *ug/kg*, *mg/kg*, *g/kg*. In order to keep only trials of a similar nature in the dataframe, so that their bioactivity values can present a pattern that correlates with features, it is necessary to define **inclusion and exclusion terms**. Filling in the **Assay Included Terms** and **Assay Excluded Terms** fields will then depend on careful curation of terms present in the **assay_description** column. If you consider the description obtained through the Assay description count button insufficient, you can make a new request to the ChEMBL Web Client using the **Explore by Assay** button, which will provide a different description in the **description** column, allowing the selection of other terms.

With the *Target Type*, *Organism Name*, *Target ChEMBL ID*, *Assay Type*, *Assay Metric*, and *Assay Unit* fields filled in, it will be possible to generate the base dataframe **df2_base_<Target ChEMBL ID>_<Organism_name>.csv** in the INTERNAL_DATA folder. Based on the **Molecule ChEMBL ID** or **Canonical SMILES**, it is possible to search for specific information about a given compound. The lower part allows you to generate bar graphs with the relative frequencies for a given column. The **update** button allows you to update the dataframe options in the INTERNAL_DATA folder for the **internal data list** combobox. When you select the dataframe from this list, the **Columns** combobox will be updated, and then you just need to choose the column and click the **View Graph** button to view it (Figure 6).

Step 2 - Preprocessing and exploratory analysis: In this step, the following preprocessing tasks are performed: Selection of columns of interest in the base dataframe, removal of instances with empty or null values, conversion of units, treatment of repetitions, normalization/transformation of bioactivity values, and removal of outliers. At the end of each block, if the dataframe selection option does not open automatically, it is important to select the last dataframe generated with the **SelectDataFrame** button (Figure 7).

Figure 7. Window for step 2 of Preprocessing and Exploratory Analysis.



Step 3 - Preprocessing and Statistical Analysis: In this step, the data can be categorized as **Active**, **Intermediate**, and **Inactive** according to their Bioactivity values (MIC, IC50...) considering the first quartile (Active if less than or equal) and the third quartile (Inactive if greater) as reference values. Next, we can generate columns with Lipinski's properties (**Set Lipinski Descriptors**) and filter the compounds into specific value ranges according to the rules of drugability (Filter by Lipinski's Rule). As mentioned earlier, at the end of each block, it is important to select the last dataframe generated with the **SelectDataFrame** button. This will display the classes in the **Select groups** multiple selection box. There are two types of statistical analyses: for class comparison and for variable correlation. In the first, it is necessary to evaluate the assumptions of normality and homoscedasticity by selecting the comparison variable in **Select Variable** and clicking the **Verify Assumptions** button (**Figure 8**).

Figure 8. Window for step 3 of Preprocessing and Exploratory Analysis.

In select groups, if only two classes (Active and Inactive) have been chosen, the **One-Way ANOVA** and **Kruskal-Wallis Test** analyses will be disabled (used for three or more groups). If the selected variable does not show normality and homoscedasticity, the nonparametric **Mann-Whitney U Test** should be used instead of the **Student's t-Test**. Clicking on the class comparison method button will generate a text window with the **interpretation of the results** and, at the end, a **density graph** (**Figure 9**).

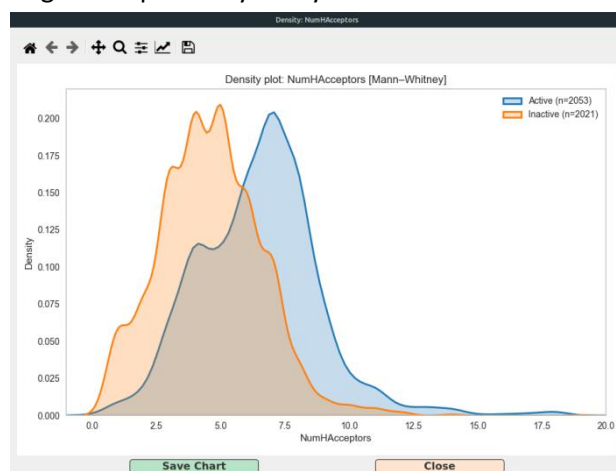


Figure 9. Density graph visualization window for class comparison.

To understand how correlated the variables are and how much the association between them is able to differentiate the groups (Active and Inactive), we can select a new dataframe in Select DataFrame. Then we choose the variables. **The Pearson** (parametric) and **Spearman** and **Kendall** (non-parametric) **correlation** tests will evaluate the variables selected in **Select Variable 1** and **Select Variable 2**, generating an interpretation window and then the linear curve graph with confidence interval. Different types of trend lines can be plotted (moving average, 2nd degree polynomial, LOESS/LOWESS, splines, and GAMs). The **Mann-Kendall Trend Test** button will provide the type of trend (whether monotonically positive or negative) and the Slope value in the boxes below (This gives an idea of how much the value of variable 1 is changing per unit of variable 2, i.e., the slope of the linear correlation curve). Clicking the **View Scatter Plot** button will generate a 2D or 3D graph according to the option selected in the corresponding checkbox (**Figure 10**).

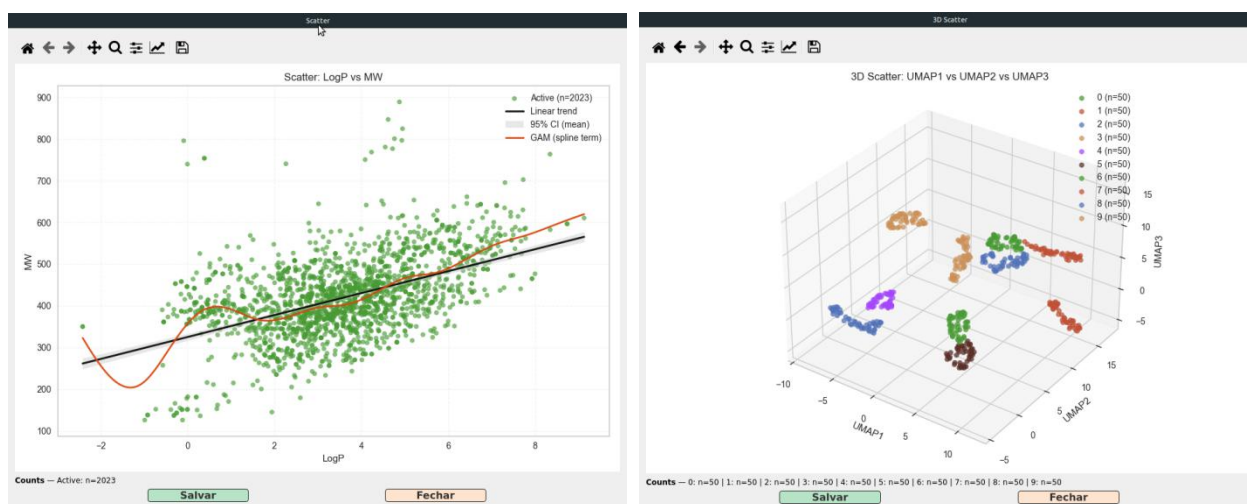


Figure 10. Windows with 2D and 3D Scatter Plot graphs for comparing variables and differentiating classes.

Step 4 - Feature engineering: In this step, the main descriptors that will compose the set of predictor variables (X values) will be generated and treatments for dimensionality reduction will be applied. In **Descriptors builder**, the checkbox for the type of descriptor (**1D/2D, 3D, and Fingerprint**) to be listed in **Select Descriptors** must be checked. More than one descriptor type can be chosen from the list. Next, select the column containing the structural information (canonical smiles) that will be used to generate the descriptors, the column corresponding to the response variable (MIC, IC50, Potency, etc.), and the column that identifies each compound (such as molecule_chembl_id). Other processes that can be selected in the checkbox before generating the descriptors are: Removal of salts, detection of aromaticity, standardization of tautomers, standardization of nitro groups, maintenance of 3D coordinates (if the structures are already in 3D format) or Conversion to 3D, if the structures do not have 3D coordinates and a 3D descriptor has been selected (**Figure 11**).

Descriptors can be calculated directly by clicking the **Generate Descriptors** button or by opening the PaDEL-Descriptor interface by clicking the **Open-GUI PaDEL-Descriptor** button. To reduce dimensionality, select: bioactivity column, class column (used only if the task type selected in the CONFIG tab is classification), and the range of indices corresponding to the descriptors. When you click **Select DataFrame** to choose the dataframe containing the descriptors that were recently calculated, the program automatically adds the names of the corresponding columns. It is always important to check that the index range in Features Columns Range actually corresponds only to the descriptors, considering that the first column has index 0 and thus the index of the last column corresponds to the total number of



columns minus one. From then on, the scaling, selection, and projection processes can be selected and applied. However, if you want the program to recommend the most appropriate options for the nature of your descriptors, you can also define the **Feature Type** (continuous, binary, categorical, or ordinal), **Attribute Option** (with noise/outliers, sparse, dense, low cardinality, high cardinality, high correlation, or non-linear correlation), and the **Model Type**, which defines the type of AI model you want to apply to the data (tree, linear, distance, probabilistic, neural network, or clustering).

The screenshot displays the 'Features Engineering' window of the CODRUG software. The interface is divided into several sections:

- Select DataFrame:** A dropdown menu showing 'df9_descriptors_CHEMBL360_Mycobacterium tuberculosis.csv' and a 'Random State' field set to 123.
- Descriptors Builder:** Includes checkboxes for '1D/2D' and '3D' (checked), and 'Fingerprints' (checked). A list of descriptors is shown, with 'ECFP6 (RDKit)' selected. Other options include 'Select Structure Column', 'Select Bioactivity Column', 'Select Name Column', and checkboxes for 'Remove salt', 'Detect Aromaticity', 'Standardize Tautomers', 'Standardize Nitro Groups', 'Retain 3D coordinates', and 'Convert to 3D'. Buttons for 'Generate Descriptors' and 'Open GUI PaDEL-Descriptor' are present.
- Dimensionality Reduction:** Includes fields for 'Label Column' (MIC), 'Class Column' (Class), and 'Feature Columns Range' (1 to 881). It also has 'Features Types' (Continuous Numeric), 'Attribute Options' (Noisy/Outliers), and 'Model Type' (Tree/Ensemble (RF, XGB...)). Recommended scaling is 'MinMaxScaler', selection is 'Elastic Net', and projection is 'UMAP'. Projection parameters are 3, 0.9, 15, 0.7, with PC1: 16.7%, PC2: 12.4%, PC3: 9.7%.
- Manipulate Dataframes:** Includes fields for 'Select DataFrame 1' and 'Select DataFrame 2', and 'Select Index 1' and 'Select Index 2'. It has checkboxes for 'By Rows' and 'By Columns' (checked), and buttons for 'Merge from Dataframes', 'Remove from Dataframes', and 'Delete Dataframes'.
- Tensor Builder:** Includes fields for 'Framework' (TensorFlow), 'Device' (cpu), 'Features dtype' (float32), and 'Label dtype' (auto). It also has 'Feature Range' (1 to 881), 'Label Column' (MIC), 'Feature Column', 'Weight' (1.000), 'Sample Weight Col', and checkboxes for 'Return Label', 'Use Columns Weights', 'Apply after scaling', 'Return Meta', 'Save Weights in Meta', and 'Apply to all dummies'. A 'Generate Tensor' button is also present.

Figure 11. Window for step 4 of feature engineering and dimensionality reduction.

In the **Manipulate Dataframes** section, we can make some adjustments by merging or removing columns or rows between dataframes or even deleting the dataframe completely. In the **Tensor Builder** section, it is possible to generate a one-dimensional vector that can be used as input for artificial neural networks (ANN). These models will be implemented later.

Step 5 - Similarity and Applicability Domain Analysis: In this step, we must select the internal dataframe by clicking **Select Internal DataFrame**. This dataframe must be the same one used for training and testing the ML models, but containing only the names column and the descriptor columns, i.e., without the Bioactivity and Class columns. Similarly, the external dataset, i.e., the set of compounds to which the ML model will be applied to predict the response variable (Bioactivity), must be selected by clicking **Select External DataFrame** (Figure 12). These two dataframes can be prepared and adjusted in the previous step using the **Manipulate Dataframes** session. The parameters k (for KNN), α , and Fingerprint are already predefined. Clicking on Compute AD will generate a table containing the results of **leverage, knn, Mahalanobis, and Tanimoto**, ultimately generating a score and a verdict: **Within AD, Borderline, Outside AD** (Figure 13).



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Menu Help CPU: 0 %

HOME CONFIG STEP 1 STEP 2 STEP 3 STEP 4 STEP 5 STEP 6

Applicability Domain and Similarity Analysis

Select Internal DataFrame Internal_data.csv

Select External DataFrame df_external_descriptors_.csv

Set AD Parameters:

k (kNN): 5

α for χ^2 cut (MD): 0.950

Fingerprint: Morgan ECFP6

☐ Project PCA for plots

Compute AD

Plot Williams

Hist Mahalanobis

PCA Scatter (AD)

Similarity Dist

AD: 25%

<< BACK NEXT >>

Figure 12. Window for step 5 of Similarity and Applicability Domain Analysis (AD).

Visualização do CSV Selecionado

		leverage_cut	leverage_ok	mahal	mahal_cut	mahal_ok	knn_mean_dist	knn_cut	knn_ok	notok	notok	ad_score	ad_verdict
1	1	0.5636346964973409	False	86.07739555817851	46.415464431749335	False	58.01540302058818	54.37050177074907	False	...	...	25	Outside AD
2	3	0.5636346964973409	True	49.73509685399713	46.415464431749335	False	33.70325781121663	54.37050177074907	True	...	...	75	Within AD
3	2	0.5636346964973409	False	94.02201022335248	46.415464431749335	False	62.328693014409836	54.37050177074907	False	...	...	25	Outside AD
4	9	0.5636346964973409	False	72.4242705081354	46.415464431749335	False	47.40873564292492	54.37050177074907	True	...	...	50	Borderline
5	1	0.5636346964973409	False	67.19014019122231	46.415464431749335	False	48.12738581999524	54.37050177074907	True	...	...	50	Borderline
6	1	0.5636346964973409	False	83.46827684507747	46.415464431749335	False	55.31330897942795	54.37050177074907	False	...	...	25	Outside AD
7	8	0.5636346964973409	False	73.1287285801007	46.415464431749335	False	47.21067385926359	54.37050177074907	True	...	...	50	Borderline
8	2	0.5636346964973409	False	72.63527078990602	46.415464431749335	False	49.369144995085065	54.37050177074907	True	...	...	50	Borderline
9	9	0.5636346964973409	False	80.77758624371747	46.415464431749335	False	54.0084048940029	54.37050177074907	True	...	...	50	Borderline
10		0.5636346964973409	False	74.10003405996369	46.415464431749335	False	52.361756588799814	54.37050177074907	True	...	...	50	Borderline
11	06	0.5636346964973409	True	43.74989986734951	46.415464431749335	True	29.285602262918037	54.37050177074907	True	...	...	100	Within AD
12	1	0.5636346964973409	False	76.58502752265883	46.415464431749335	False	51.17210911815157	54.37050177074907	True	...	...	50	Borderline
13	2	0.5636346964973409	False	67.52539901992371	46.415464431749335	False	45.768698934817834	54.37050177074907	True	...	...	50	Borderline
14	07	0.5636346964973409	True	47.49151405657627	46.415464431749335	False	35.237373282977096	54.37050177074907	True	...	...	75	Within AD
15	4	0.5636346964973409	False	101.6018444366192	46.415464431749335	False	65.23822547522045	54.37050177074907	False	...	...	25	Outside AD
16	8	0.5636346964973409	False	74.64111801524471	46.415464431749335	False	51.955454155916144	54.37050177074907	True	...	...	50	Borderline
17	5	0.5636346964973409	False	99.71063231984934	46.415464431749335	False	66.99417989615054	54.37050177074907	False	...	...	25	Outside AD
18	7	0.5636346964973409	False	99.48937101830398	46.415464431749335	False	64.01186004556796	54.37050177074907	False	...	...	25	Outside AD

Salvar Fechar Rows: 15500 Columns: 15

Figure 13. Window of step 5 of Compute AD results.



Step 6 - Machine Learning Models: Screening and application. In this step, the models will be screened and ranked according to the chosen performance metric. The best model can be saved and used for application in an external dataframe containing the same descriptors. First, we must choose the internal dataframe that will be used to rank the models by clicking the **Select Internal DataFrame** button and leaving the **Internal DataFrame** checkbox enabled. The **External DataFrame** checkbox will only be checked when we use the **Load** button to load a saved model and thus apply it to the external dataframe (**Figure 14**).

In the vertical **Setup** tab, under Set Setup Parameters, the column corresponding to the response variable must be selected (**Select Y Column**). The other parameters can be enabled and adjusted or left unchecked so that they are applied automatically. The **ignore_features** checkbox must be checked and the columns that will be ignored for the model must be described, such as **Name** and **Class** (if it is a regression model) or the **Bioactivity** column (if it is a classification model). Click the **Setup** button to generate the parameter table and the **USI** screening identifier code that will be used to name the results folder.

In the vertical **Compare** tab, check the **sort** checkbox to define the metric that will be used to rank the models (usually R2 for regression and Accuracy for classification). If you do not want this first screening to take too long, you can check the exclude model box and describe the model with the longest training time (lightgbm). Click the Compare button and wait for the progress bar until the dataframe with the ranked models and their metrics is generated. In **Evaluate**, the **Use_gpu** checkbox can be checked so that validation is performed using the GPU, speeding up the process. In **Tuning**, check the **Metric optimization** option and select the metric under which the model will be optimized with hyperparameter adjustment.

The screenshot shows the CODRUG software interface at Step 6. The main window is titled "Machine Learning Models: Screening and Application". It features a sidebar on the left with tabs: Setup, Compare, Evaluate, Tuning, and Save/Load. The main area is divided into several sections:

- DataFrame Selection:** Includes buttons for "Select Internal DataFrame" (selected) and "Select External DataFrame". The internal dataframe is "df12_projection_PCA__2025-11-24_13-32.csv" and the external is "df_external_descriptors__csv". A "Random State" field is set to 123, and a "USI" field is empty.
- Set Setup Parameters:** A table with checkboxes and dropdowns for various parameters:

Parameter	Value
Internal DataFrame	☒
External DataFrame	☐
Select Y column:	MIC
n_jobs	-1
normalize	True
transformation	True
imputation_type	simple
remove_outliers	True
feature_selection	True
use_gpu	Auto
fold	10
fold_strategy	kfold
data_split_shuffle	True
data_split_stratify	True
train_size	0.70
ignore_features	'Name','Class','ID','molecule_chembl_id','zin_id'
- Chart Selection:** Two lists of charts for "Regression" and "Classification". The "Regression" list includes pipeline, residuals, error, cooks, rfe, learning, vc, manifold, feature, and feature_all. The "Classification" list includes auc, threshold, pr, confusion_matrix, error, class_report, boundary, rfe, learning, manifold, and calibration. The "Clustering" list includes cluster, tsne, elbow, silhouette, distance, and distribution.
- Action Buttons:** A vertical column of buttons: Setup, Compare, Evaluate, Tuning, Create, Assign, Save, Load, and Predict.

Navigation buttons at the bottom are "<< BACK" and "NEXT >>". The top status bar shows "CPU: 3 %".

Figure 14. Window for step 6 of Screening and application of machine learning models.



After tuning, the best model can be saved in the `~/CODRUG/JOBS/DATE_TIME_TASK_<JOB_NAME>/MODELS` folder in .pkl format by clicking the **Save** button. This .pkl model can be loaded later by clicking the **load** button and applied to the prediction of bioactivity values for an external or class dataframe. To do this, you must first click on **Select External DataFrame**, check the External dataframe checkbox in Set Setup Parameters, redefine the **select Y Column**, and **ignore_features**, if necessary, and execute the **setup** button. Only then can the load be performed and the predict button be used to generate a new column of predictions of Y values based on X values (Descriptors). Depending on the type of task selected, the **Chart Selection** list will be enabled or disabled. After setup, comparison, and tuning, you can generate training, testing, and validation graphs for the best model. They can be selected in the respective selection box and generated by clicking the **Plot Model** button. These selected graphs will be saved in the folder `~/CODRUG/JOBS/DATE_TIME_TASK_<JOB_NAME>/RESULTS/USI/<USI code>/MIDIA`. Some examples are shown in **Figures 15 to 20** below.

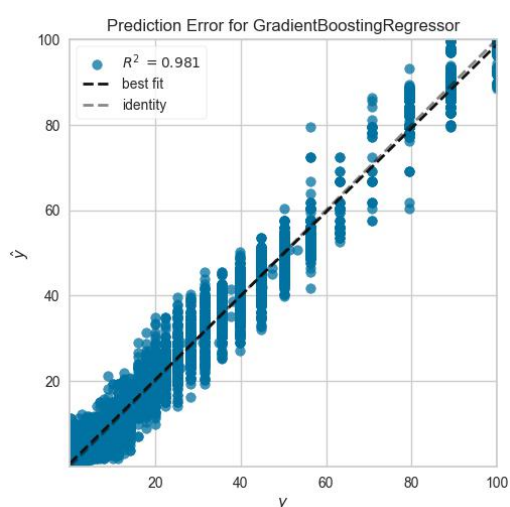


Figure 15. Prediction error graph in a Regression task.

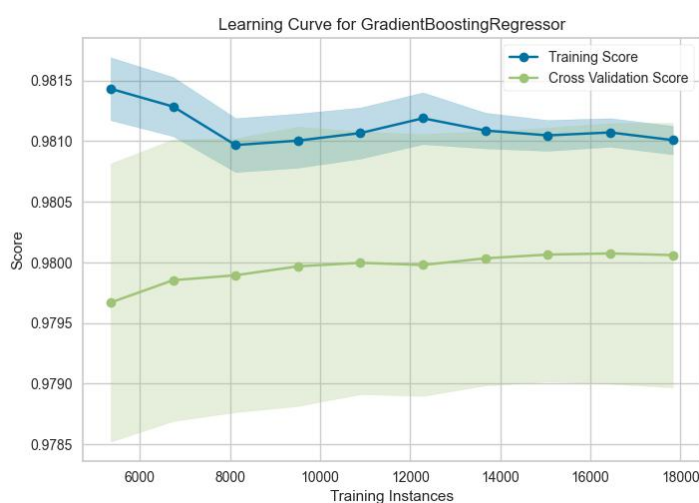


Figure 16. Learning Curve graph in a Regression task.

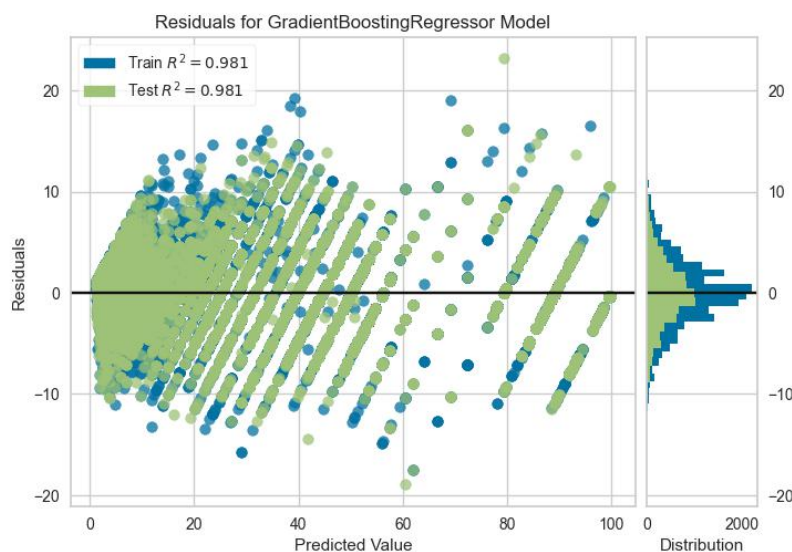


Figure 17. Residual graph in a regression task.

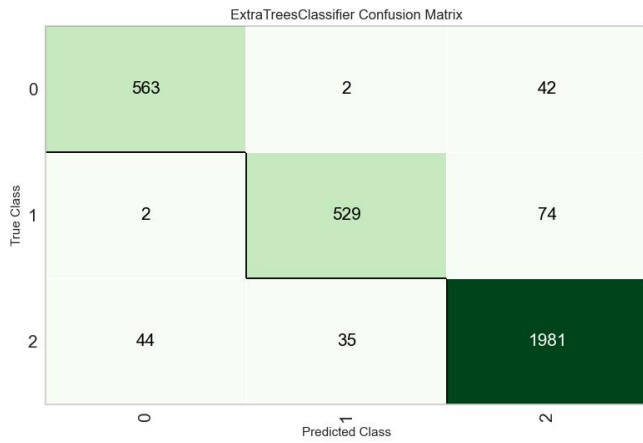


Figure 18. Confusion Matrix graph in a Classification task.

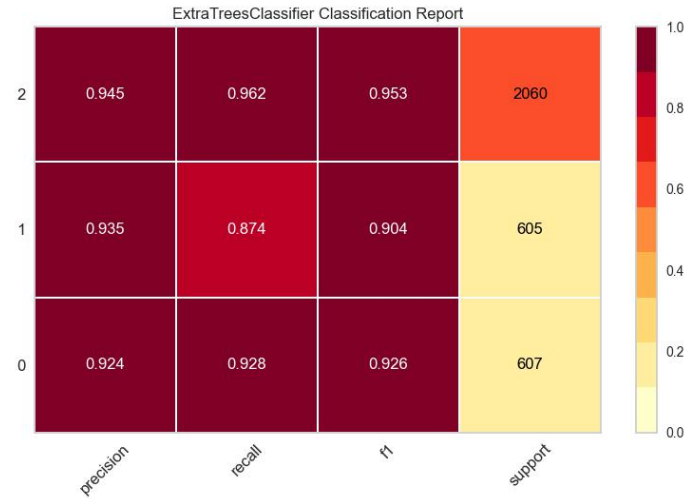


Figure 19. Classification report.

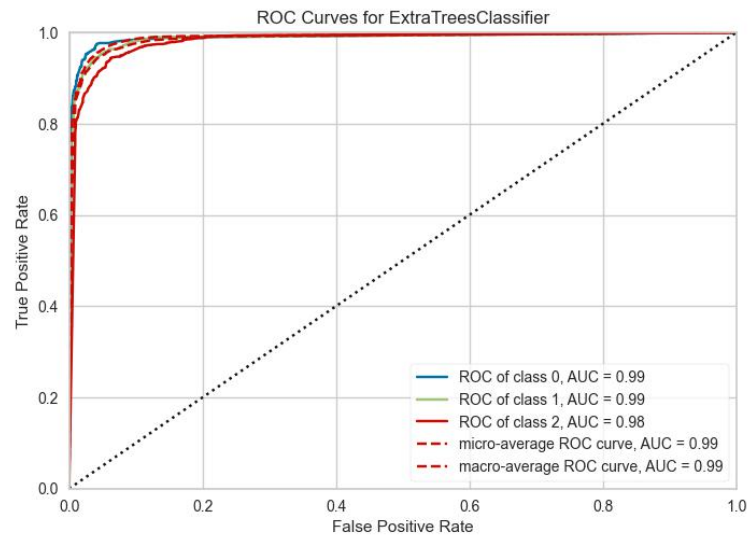


Figure 19. Area Under the Curve (AUC) graph in a Classification task.

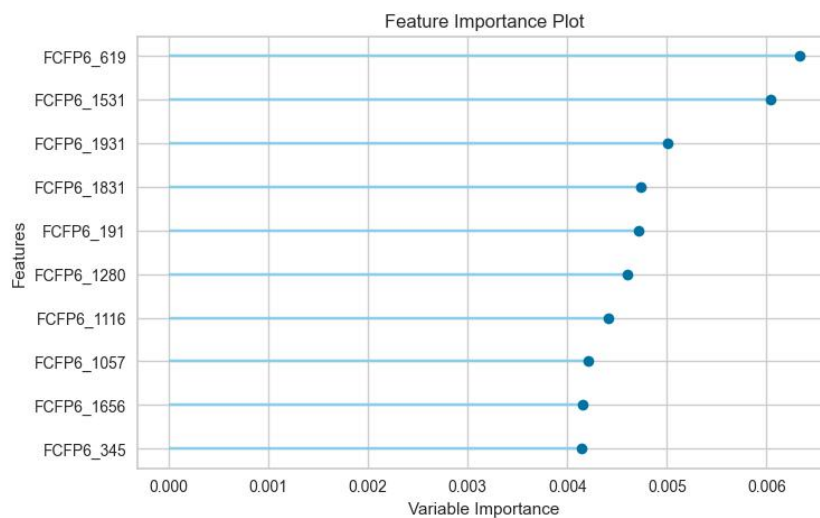


Figure 20. Feature Importance graph in a Classification task.